Predicting flight departure delays

Project Description and Data Preparation Report

1. Project overview

Our project will study flight departure delays and build a model that estimates whether a flight will leave at least **15 minutes later than scheduled**. We will use historical flights from New York's three main airports: Newark (EWR), John F. Kennedy (JFK) and LaGuardia (LGA).

The main question is: **can a flight's schedule, airline and route help predict its risk of departure delay?** We will also investigate how delay frequency varies with departure time, day of the week, month and origin airport.

The model will use information available before departure. Its output will be an estimated probability of a delay and a delayed/not-delayed prediction. This will be a historical case study, rather than a live flight-tracking service.

2. Data we will use

We will use the public **nycflights13** dataset, maintained by Hadley Wickham. It contains **336,776 flight records from 2013**, with flight data originating from the US Bureau of Transportation Statistics. We will use three related tables:

Table	Contents	Use in our project
flights	Schedules, airlines, routes, distances and observed delays.	Main dataset for analysis and prediction. Each record represents one flight.
airlines	Airline codes and names.	Replace codes with readable names in charts and comparisons.
airports	Airport codes, names and locations.	Describe routes and identify airports in visualisations.

These tables can be downloaded directly, so the project does not depend on scraping airline websites or accessing a paid API.

3. Planned approach and expected outcome

We will first explore the data with charts and summary statistics, then compare a decision tree and a Random Forest with a simple baseline. Python, pandas, Matplotlib and scikit-learn will support this work. The final result will combine a delay prediction model with visual explanations of the patterns found.

The project applies the course topics on data collection and integration, preprocessing, exploratory analysis, visualisation, machine learning and evaluation. The dataset is manageable on a standard computer.

Data preparation and use

4. Selected information and its role

Information	Dataset fields	How we will use it
Flight date	year, month, day	Derive month and weekday to study calendar patterns.
Scheduled departure	sched_dep_time	Extract the local departure time and study hourly patterns.
Airline	carrier	Compare airlines and provide a categorical model input.
Route	origin, dest	Compare airports and routes; use their codes as model inputs.
Flight distance	distance	Use the route distance in miles as a numerical model input.
Observed departure delay	dep_delay	Create the outcome to predict: delayed if the value is at least 15 minutes.

The observed delay is used to label historical flights, **not as a prediction input**. Actual departure and arrival times, arrival delay and time spent in the air will also be excluded from the inputs, since they would not be known when making a prediction before departure.

5. How we will prepare the data

Combine and check. We will link the flight table to airline and airport metadata using their codes, check for duplicate records and missing values, and verify that dates, times and numerical values are consistent. Flights without an observed departure delay will be excluded from supervised learning. Missing airport names or locations will not require removing an otherwise usable flight.

Create model inputs. We will derive the weekday and useful time features from the scheduled date and departure time. Airline and airport categories will be encoded into numerical inputs. The target will have two classes: a departure delay of at least 15 minutes, or a smaller delay/early departure.

Separate learning and evaluation. We will split flights chronologically: earlier records for training, a later period for choosing model settings, and the final period for testing. Encoding will be fitted only on the training data. We will compare predictions with observed outcomes using a confusion matrix, precision, recall and F1 score.

The initial model will use schedule information; weather and real-time operational data are outside the initial scope.

Data source: nycflights13.tidyverse.org (CC0). Variable definitions: Flights data documentation.